

Mind the Gap: An Energy-Aware Adaptive Cruise Controller for an Electrified Off-Road SUV

Sponsors: [Faculty Sponsor], Department of Electrical Engineering and Computer Science; [EcoCAR Faculty Advisor], UTK EcoCAR Team (liaison to the Propulsion Controls and Modeling swimlane)

Computer Engineering, Electrical Engineering (CS secondary) • Propulsion Controls & Modeling • Builds on the PCM Tool Check (Nov 5, 2026)

Project Background and Objective

On the Stellantis track of this DOE-sponsored, Argonne-managed competition, the team integrates a rear electric drive unit and a team-built high-voltage battery into a 2026 Jeep Cherokee, then writes the software deciding how torque is split across the engine, the front electrified powertrain, and the new rear EDU. Adaptive Cruise Control is the entry point to automated driving and an unusually clean senior design problem: a closed-loop controller with a clear safety envelope, a measurable objective, and a real vehicle network to talk to. It is also open territory — the competition’s GM track has a full automated-driving swimlane and the Stellantis track has none, making a working ACC feature a real differentiator and a strong IEEE ITEC paper candidate. The project designs, simulates, and bench-demonstrates an adaptive cruise controller with an energy-efficiency objective layered on conventional car-following. The headline result is how much energy an “eco-ACC” gap policy saves over ordinary cruise control.

Project Scope

- **Network characterization.** Read-only CAN logging, decode against the signal database, identify longitudinal-control signals, and measure periods, latency, jitter, and bus load to fix the achievable loop rate.
- **Requirements and safety envelope.** Set-speed and gap policy, acceleration and jerk limits, driver override, degraded modes; preliminary hazard analysis in the ISO 26262 style the VSE swimlane already uses, fully traced.
- **Controller design.** Time-gap spacing policy with cascaded range and speed loops feeding a torque request; compare a tuned PID design against model predictive control; analyze string stability.
- **Energy-aware layer.** Shape the speed profile inside the gap window, maximize regenerative braking, and bias torque between the front powertrain and the rear EDU.
- **Verification and demonstration.** MIL simulation against the MathWorks vehicle model on EPA cycles plus cut-in, lead-braking, target-loss, and grade scenarios; then closed loop over physical CAN against a simulated lead vehicle.

Technical Considerations

- **Actuation is out of scope by design.** Vehicle interaction stays read-only and no modification is needed. The track permits cruise control but prohibits direct engine control — torque requests only. Confirm scoring treatment with organizers; the value is differentiation, not points.
- **Sensor access is the largest risk.** Stock radar and camera data may sit on an unreachable bus, so the controller interface is defined independently of the sensor source and simulated object lists are an acceptable input.
- **Timing is measured, not assumed.** A loop over a shared bus inherits its jitter. Safety behavior — override, brake precedence, sensor dropout, graceful degradation — is where the judgment lives, and efficiency that costs drivability is not shippable.

Deliverables

- Requirements specification and preliminary hazard analysis with traceability matrix; CAN interface control document with measured timing.
- Version-controlled Simulink/Stateflow model with generated C code and documented architecture.
- MIL results across the scenario library, including the energy comparison against conventional cruise control.
- Bench demonstration with recorded video; report and presentation structured for adaptation into an IEEE ITEC paper.

Assumptions and Dependencies

Requires MATLAB, Simulink, and Stateflow, the MathWorks EcoCAR vehicle model, a CAN interface (Vector, Kvaser, or PEAK with `python-can/cantools`), signal database definitions, and a bench that can close the loop over CAN. Builds on the toolchain from the PCM Tool Check. All simulation and bench work proceeds independently if the vehicle is delayed. Students need MATLAB/Simulink, C, Python, and feedback control; ECE 4xx controls coursework is the natural prerequisite.

Support and Guidance

Mentorship on control design, embedded software, and vehicle networks from the faculty sponsor, with requirements and verification guidance from the EcoCAR advisor and swimlane leads. The competition provides scheduled reviews with Stellantis, MathWorks, and Argonne engineers, giving the project real external deadlines. Biweekly sponsor meetings plus EcoCAR checkpoints.